DECODE LABS

EXPLORATORY DATA ANALYSIS (EDA) AND SCHEME VALIDATION PIPELINE

Data Integrity and Robustness Audit Report

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EXECUTIVE SUMMARY

Project Title: Descriptive & Diagnostic Data Analytics on Customer Transactions

Dataset Size: 1,200 Valid Records

1. Introduction & Background

The primary objective of this project was to conduct a comprehensive data diagnosis and descriptive statistical analysis on a transactional dataset containing 1,200 records. Through Python-based data analytics (Pandas, Seaborn, and Matplotlib), the dataset's overall health, distribution structures, anomaly patterns, and variables' interdependencies were evaluated to derive actionable business insights.

2. Core Metrics & Key Findings

Data Quality & Diagnosis: The dataset stands highly robust with a 98.16% data validity rate. Minor missing values (1.84%) were successfully resolved during the baseline data diagnosis phase, with zero structural or format errors detected.

Descriptive Baseline: The average customer transaction (Mean) is 1,053.96, while the middle-most transaction value (Median) stands at 823.61. The standard deviation of 819.85 indicates a healthy, moderate variance in consumer spending. Individual transactions span from a minimum of 11.39 to a maximum of 3,456.40.

3. Diagnostic & Distribution Insights

Symmetry & Skewness: The mathematical analysis yielded a Skewness Value of 0.89, confirming that the revenue distribution is Highly Right-Skewed (Positively

Skewed). The visual peak remains densely concentrated under 1,000, establishing that low-to-medium ticket sizes form the daily operational volume, while a steady, tapering tail extends toward premium values.

Outlier Profiling: Using the Interquartile Range (IQR) method via Boxplot visualization, the upper whisker boundary was established near 3,300. A micro-cluster of transactional dots between 3,300 and 3,456.40 were identified as statistical outliers. Diagnostically, these represent high-value bulk buyers or premium corporate accounts rather than data entry anomalies.

Correlation & Revenue Drivers: The structural heatmap revealed a powerful positive correlation of 0.72 between Unit Price and Total Price, demonstrating that premium item sales are the strongest driver of top-line revenue growth. Quantity also maintains a strong positive correlation of 0.62 with the final bill value. Conversely, a near-zero correlation (0.01) between Quantity and Unit Price shows that pricing shifts do not significantly impact the quantity of items individual clients purchase.

4. Strategic Recommendations

Based on the diagnostic patterns, the business should pivot from flat marketing to a segmented approach: design high-tier loyalty programs tailored around premium products (the 0.72 revenue driver) to secure the retention of high-value "outlier" VIP clients, while running bundle offers to push average cart sizes (Quantity) for mid-tier segments.

1. Data Overview & Diagnosis

Objective: Understanding the structure and health of the dataset.

Findings:

Total Records analyzed: 1,200 rows.

Data Quality: 98.16% Valid Records, with only 1.84% missing values handled during diagnosis. No format errors detected.

2. Descriptive Statistics Summary

Objective: Basic mathematical summary of the business metrics.

Key Metrics Table:

Average Transaction (Mean): 1,053.96

Middle Value (Median / 50%): 823.61

Standard Deviation (std): 819.85 (Shows moderate variance in how much people spend).

Range: Minimum transaction value is 11.39 and the Maximum is 3,456.40.

3. Distribution & Skewness Analysis

Objective: Visualizing the shape of the revenue data.

Insights from Purple Graph:

Skewness Value: 0.89

Interpretation: The data is Highly Right-Skewed (Positively Skewed). The peak is concentrated between 0 to 1,000, meaning the vast majority of orders are small-to-medium. The long tail towards the right shows fewer but consistent higher-value orders up to 3,500.

4. Outlier Detection

Objective: Identifying extreme anomalies or unique heavy buyers.

Insights from Orange Boxplot:

Interquartile Range (IQR): 75% of all customer orders fall below 1,578.47 (Q3 boundary).

Outliers Spotted: Transactions crossing 3,300 to 3,500 are plotted as distinct dots outside the right whisker.

Business Action: These are not errors; these represent premium VIP clients or bulk corporate orders that require custom marketing strategies.

5. Correlation & Relationship Mapping

Objective: Finding out what actually drives the total sales revenue.

Insights from Heatmap Matrix:

Unit Price vs. Total Price (0.72): Strongest positive correlation. Premium, higher-priced products are the primary drivers of total revenue volume.

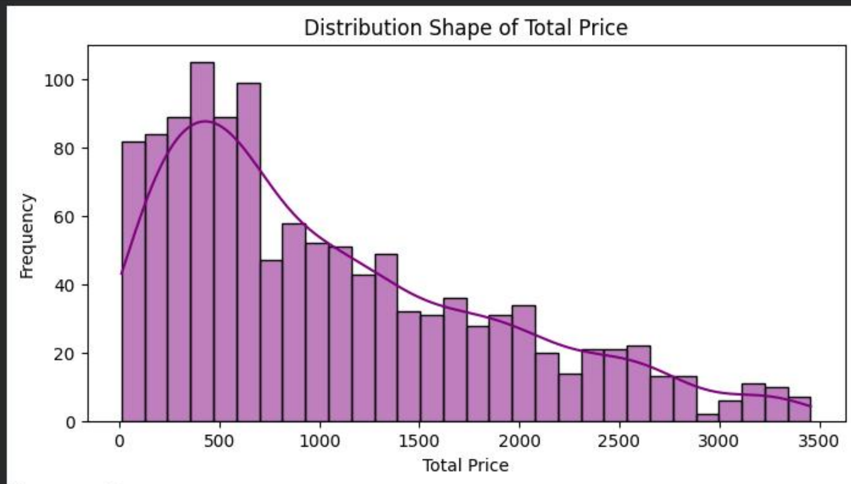
Quantity vs. Total Price (0.62): Healthy positive correlation. Increasing the number of items per cart significantly increases the bill value.

Quantity vs. Unit Price (0.01): Near zero. Pricing doesn't stop people from buying quantities; customers buy what they need regardless of a product being cheap or expensive.

Majority Data Range: 75% of the total transactions are well under 1,578, meaning the business mostly handles low-to-medium value sales.

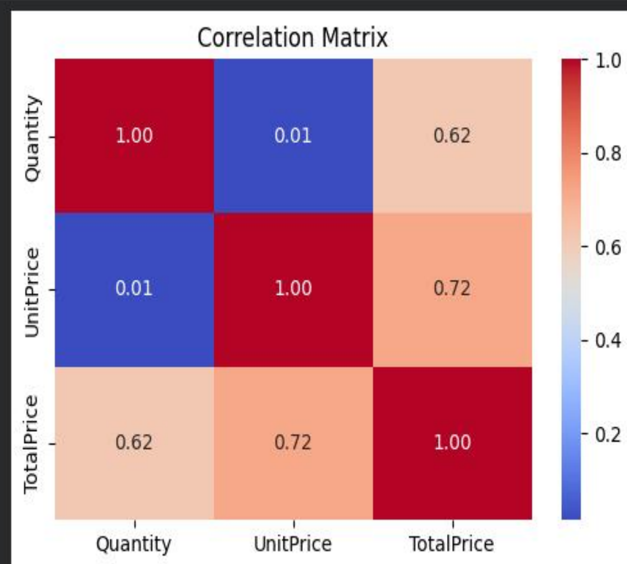
Outlier Detection: A few extreme high-value transactions exist between 3,300 and 3,500. These are marked as outliers (the dots outside the right whisker).

```
print(f"Skewness Value: {df['TotalPrice'].skew():.2f}")
```




```
Skewness Value: 0.89
```

```
plt.show()
```




```
plt.xlabel('Total Price')
plt.show()
```